

PSYCHIATRIC DISORDERS IMPACTING CHILDREN

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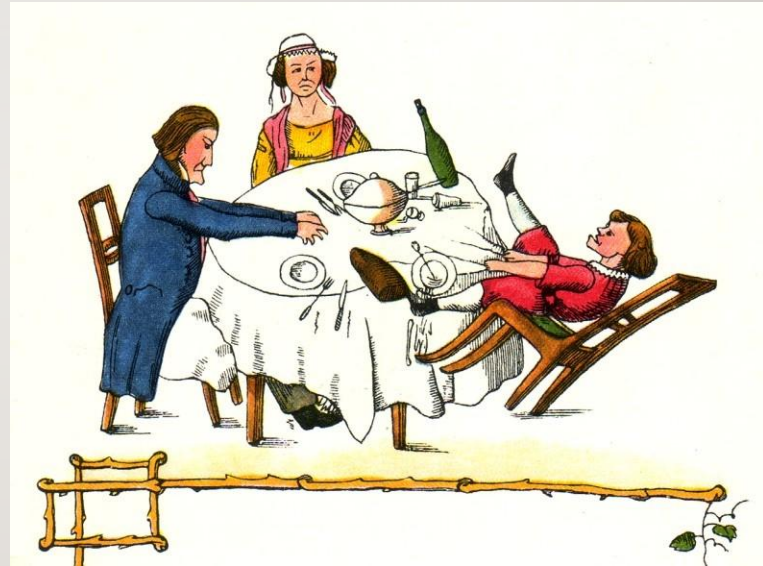
None

FINANCIAL DISCLOSURES

FIDGETY PHILLIP

DR. HEINRICH HOFFMANN, 1844

Let me see if Philip can
Be a little gentleman
Let me see, if he is able
To sit still for once at table:
Thus, Papa bade Phil behave;
And Mamma look'd very grave.
But fidgety Phil,
He won't sit still;
He wriggles
and giggles,
And then, I declare
Swings backwards and forwards
And titlts up his chair,
Just like any rocking horse; -
"Philip! I am getting cross!"



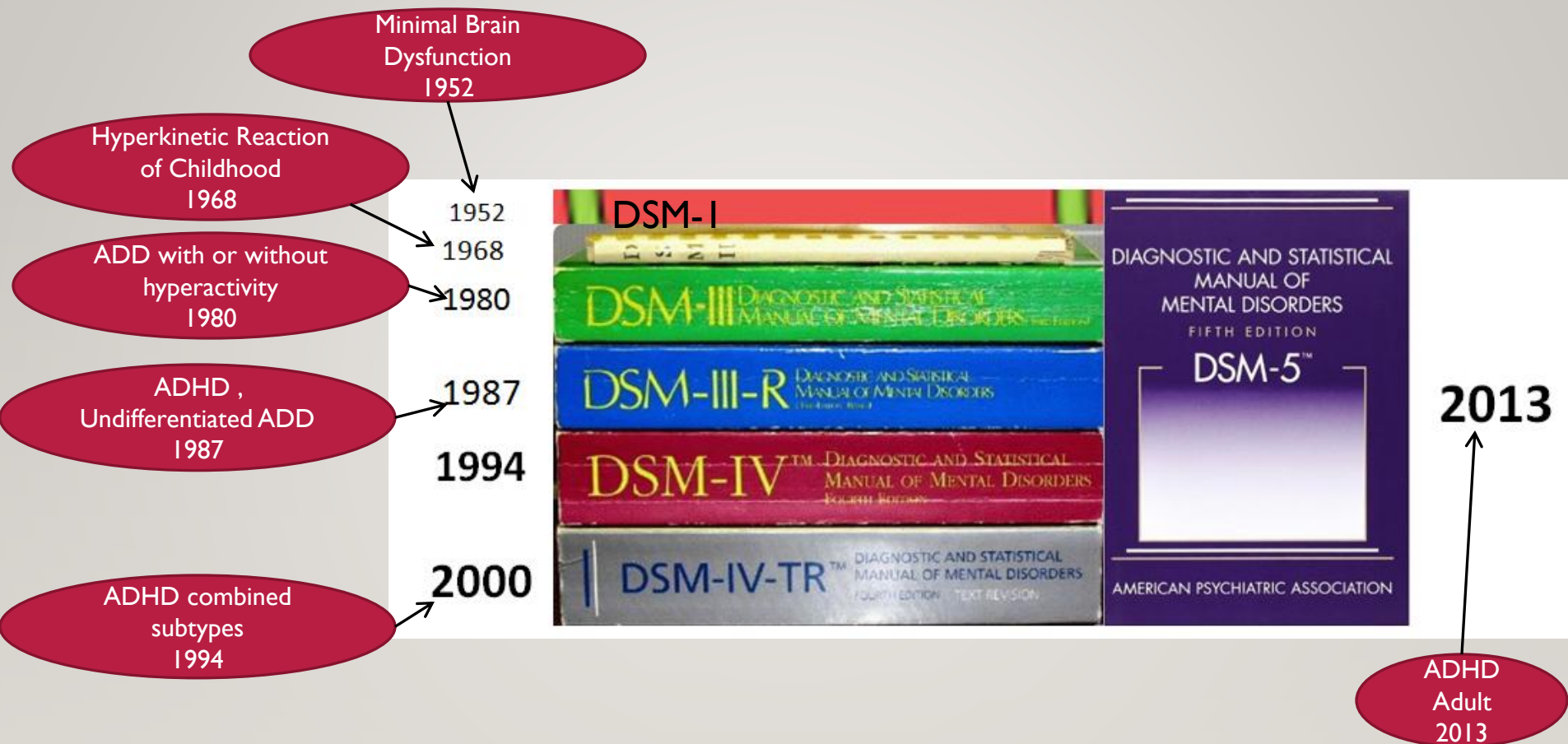
BRIEF HISTORY

MEDICAL

Year	Event
1848	Dr. Heinrich Hoffman wrote Fidgety Phil
1902	Dr. George Frederic Still first comprehensive observations in ADHD
1917-18	encephalitis epidemic Still's symptoms seen "brain damage"
1937	amphetamines were helpful in reducing hyperactive and impulsive behavior
1950s and 60s	increase in psychiatric drug intervention
1968	Dr. Stella Chess described "Hyperactive Child Syndrome"

LEGAL

Year	Event
1973	Rehabilitation Act of 1973 Section 504
2002	No Child Left Behind Act, Public Law 107-110
2004	Individuals with disabilities Education Act (IDEA)
2008	Americans with Disabilities Act Amendments Act of 2008 (ADAAA)
2009	Forest Grove School District v.T.A.



Overview of the DSM-5™ medical classification system for ADHD

- A persistent pattern of **inattention** and/or **hyperactivity-impulsivity** that interferes with functioning or development:¹
 - **For children**, six or more of the symptoms (Table) have persisted for at least 6 months to a degree that is inconsistent with developmental level, and that negatively impacts directly on social and academic/occupational activities. **Please note:** the symptoms are not solely a manifestation of oppositional behaviour, defiance, hostility or failure to understand tasks or instructions¹
 - **For older adolescents and adults (age 17 and older)**, five or more symptoms are required (Table)¹
- Several inattentive or hyperactive-impulsive symptoms present prior to age 12 years¹
- Several inattentive or hyperactive-impulsive symptoms present in two or more settings (e.g. at home, school or work; with friends or relatives; in other activities)¹
- Clear evidence that the symptoms interfere with, or reduce the quality of, social, academic or occupational functioning¹
- Symptoms do not occur exclusively during the course of schizophrenia or another psychotic disorder, and are not better explained by another mental disorder (e.g. mood disorder, anxiety disorder, dissociative disorder, personality disorder, substance intoxication or withdrawal).¹

A. A persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with functioning or development, as characterized by (1) and/or (2):

Symptoms of inattention (1)	Symptoms of hyperactivity and impulsivity (2)
Often fails to give close attention to detail or makes mistakes	Often fidgets with or taps hands and feet, or squirms in seat
Often has difficulty sustaining attention in tasks or activities	Often leaves seat in situations when remaining seated is expected
Often does not seem to listen when spoken to directly	Often runs and climbs in situations where it is inappropriate (in adolescents or adults, may be limited to feeling restless)
Often does not follow through on instructions and fails to finish schoolwork or workplace duties	Often unable to play or engage in leisure activities quietly
Often has difficulty organising tasks and activities	Is often 'on the go', acting as if 'driven by a motor'
Often avoids, dislikes or is reluctant to engage in tasks that require sustained mental effort	Often talks excessively
Often loses things necessary for tasks or activities	Often blurts out answers before a question has been completed
Is easily distracted by extraneous stimuli	Often has difficulty waiting their turn
Is often forgetful in daily activities	Often interrupts or intrudes on others

Note: The symptoms are not solely a manifestation of oppositional behavior, defiance, hostility, or failure to understand tasks or instructions. For older adolescents and adults (age 17 and older), at least five symptoms are required.

- B. Several inattentive or hyperactive-impulsive symptoms were present prior to age 12 years.**
- C. Several inattentive or hyperactive-impulsive symptoms are present in two or more settings (e.g., at home, school, or work; with friends or relatives; in other activities).**
- D. There is clear evidence that the symptoms interfere with, or reduce the quality of, social, academic, or occupational functioning.**
- E. The symptoms do not occur exclusively during the course of schizophrenia or another psychotic disorder and are not better explained by another mental disorder (e.g., mood disorder, anxiety disorder, dissociative disorder, personality disorder, substance intoxication or withdrawal).**



Specify whether:

314.01 (F90.2) Combined presentation: If both Criterion A1 (inattention) and Criterion A2 (hyperactivity-impulsivity) are met for the past 6 months.

314.00 (F90.0) Predominantly inattentive presentation: If Criterion A1 (inattention) is met but Criterion A2 (hyperactivity-impulsivity) is not met for the past 6 months.

314.01 (F90.1) Predominantly hyperactive/impulsive presentation: If Criterion A2 (hyperactivityimpulsivity) is met but Criterion A1 (inattention) is not met over the past 6 months.

Specify if:

In partial remission: When full criteria were previously met, fewer than the full criteria have been met for the past 6 months, and the symptoms still result in impairment in social, academic, or occupational functioning.



Specify current severity:

Mild: Few, if any, symptoms in excess of those required to make the diagnosis are present, and symptoms result in only minor functional impairments.

Moderate: Symptoms or functional impairment between “mild” and “severe” are present.

Severe: Many symptoms in excess of those required to make the diagnosis, or several symptoms that are particularly severe, are present, or the symptoms result in marked impairment in social or occupational functioning.

II. Epidemiology

1. Prevalence

1. Diagnosed at one point in up to 11% of school children ages 4 to 17 years old (1-2 per classroom)

1. At a given time 8.8% of school children have the diagnosis and 6.1% are on medication for Attention Deficit

Visser (2014) J Am Acad Child Adolesc Psychiatry 53(1): 34-46

2. More commonly diagnosed in boys than girls (4 - 8:1)

1. However girls are more often of the inattentive type, and may be underdiagnosed

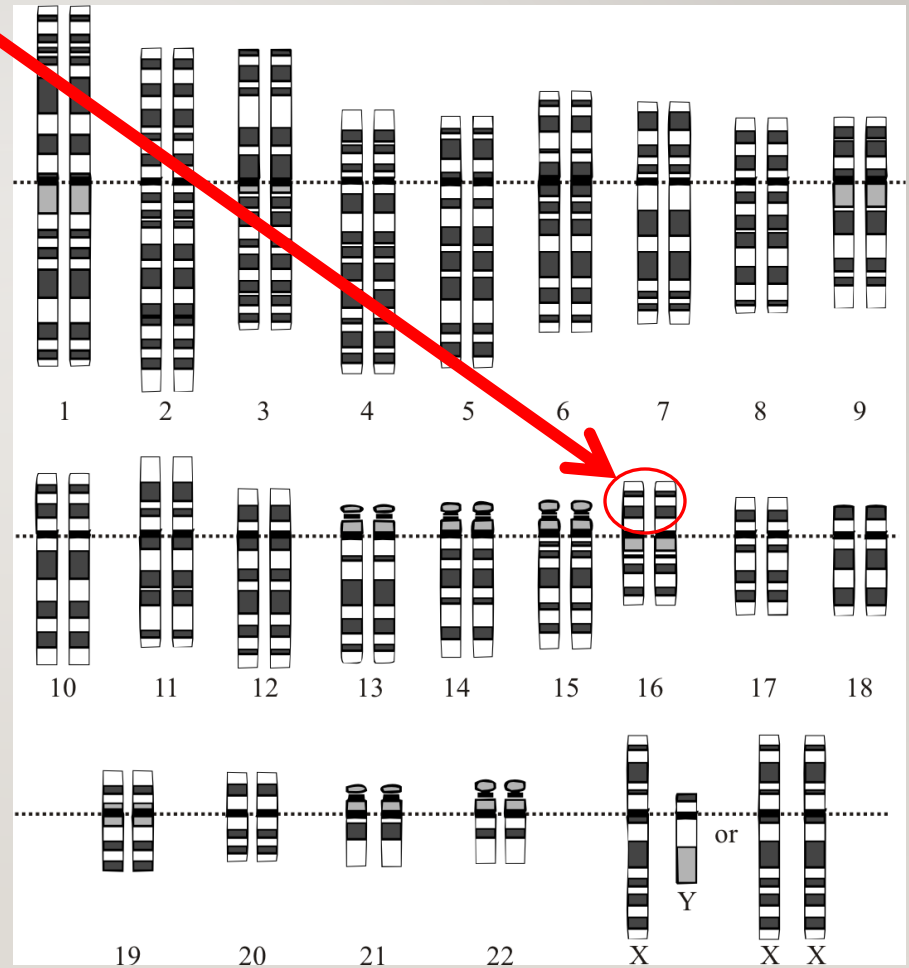
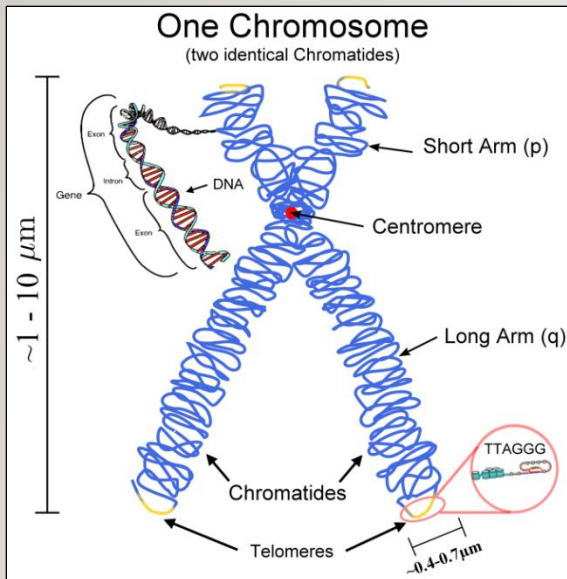
3. First degree relative with ADHD increases patient risk by 2-8 fold

2. Overlap with other Learning Disability

1. Other Learning Disability: 10% of children with ADHD
2. Learning disabled children with ADHD: 33%



Genetic Linkage of ADHD 16p13



Genes that may be linked to ADHD

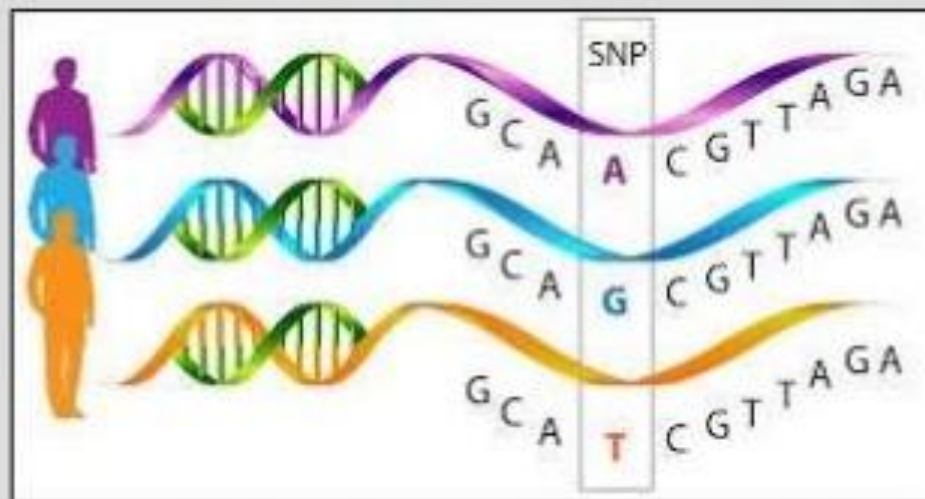
Dopaminergic neurotransmission system: DRD4, DRD5, DAT1/SLC6A3, DBH, DDC.

Noradrenergic system: NET1/SLC6A2, ADRA2A, ADRA2C).

Serotonergic system: 5-HTT/SLC6A4, HTR1B, HTR2A, TPH2.

Neurotransmission and neuronal plasticity: SNAP25, CHRNA4, NMDA, BDNF, NGF, NTF3, NTF4/5, GDNF.

SNP as Biomarker in Pharmacogenomics



- SNPs are single-letter variations in DNA sequence.
- SNPs occur every 100–300 bases along the 3 billion base pairs.
- SNP → Variant alleles → Variable Protein & Protein Functions.
- Variation in genetic information particularly due to the SNPs has been linked with the variation in drug response.

Patient Type A



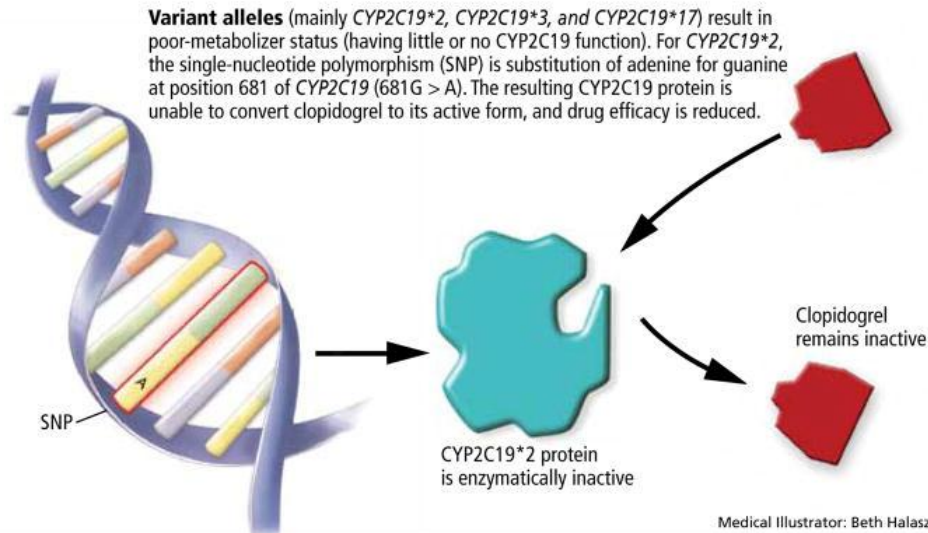
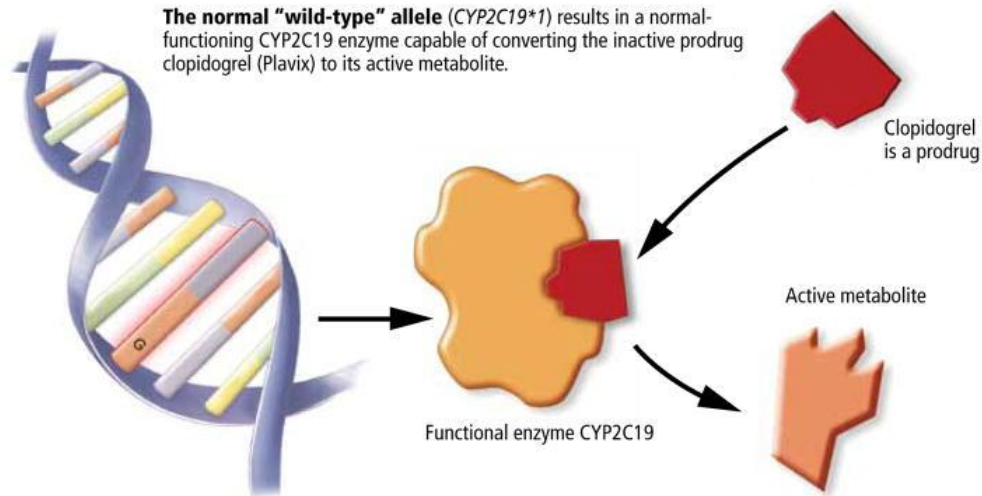
- Normal drug metabolism & response.
- Nominal side effects.

Patient Type B



- Heightened drug metabolism & response.
- Significant side effects.

Pharmacogenomic research investigates how patients' genetics influence their response to medication. Genetic variability can influence how quickly a drug is metabolized, how well a drug works, or even the likelihood of side effects. The example below depicts how genetic variation can influence the conversion of a prodrug.



KNOW THE TRUTH

MYTH

A CHILD WITH A PSYCHIATRIC DISORDER IS DAMAGED FOR LIFE.

PSYCHIATRIC PROBLEMS RESULT FROM PERSONAL WEAKNESS.

PSYCHIATRIC DISORDERS RESULT FROM BAD PARENTING.

A CHILD CAN MANAGE A PSYCHIATRIC DISORDER THROUGH WILLPOWER.

THERAPY FOR KIDS IS A WASTE OF TIME.

CHILDREN ARE OVERMEDICATED.

CHILDREN GROW OUT OF MENTAL HEALTH PROBLEMS.

TRUTH

A psychiatric disorder is by no means an indication of a child's potential for future happiness and fulfillment. Early intervention can help.

It can be difficult to separate the symptoms of a child's psychiatric disorder from a child's character. A psychiatric disorder is an illness not a personal flaw, just like diabetes or leukemia.

Parenting isn't to blame. Anxiety, depression, and learning disorders often have biological causes.

A disorder is not mild anxiety or a change in mood. Some parents resist mental health services for their children because they fear the stigma attached to diagnoses, or they don't want their kids dependent on psychiatric drugs.

Research has shown that treatment interventions have the best results during the first few years when symptoms of psychiatric disorders appear.

Skilled psychiatrists use great caution when deciding on a treatment plan that includes medication, which usually involves behavioral therapy. Medication is not the norm. Only 5% of children with psychiatric issues take medication.

Children are less likely to "grow out" of psychiatric disorders than they are to "grow in" to more disability.

1 in 10 children (aged 4–17 years) have been diagnosed with ADHD
that's **5.4 million** children in the U.S.



1 in 20 children

in the U.S. have ADHD symptoms



4.1%



of these children, will have those symptoms into the teen years.

U.S. adult population with ADHD

41.3%

of diagnosed adults have severe symptoms

that's **1.7%**



of the total U.S. adult population

CHILDHOOD MENTAL DISORDERS CAN INCLUDE:

Major Depressive Disorder Dysthymia Bipolar Disorder Attention Deficit Hyperactivity Disorder (ADHD) Oppositional Defiant Disorder (ODD) Generalized Anxiety Disorder Separation Anxiety Disorder Posttraumatic Stress Disorder (PTSD) Obsessive Compulsive Disorder (OCD) Social Phobia Specific Phobia

WHICH CAN LEAD TO:



Sexual
Risk-taking
Behavior



Substance
Use, Abuse,
or Dependence



Criminal
Behavior



Poor Coping
& Social Skills

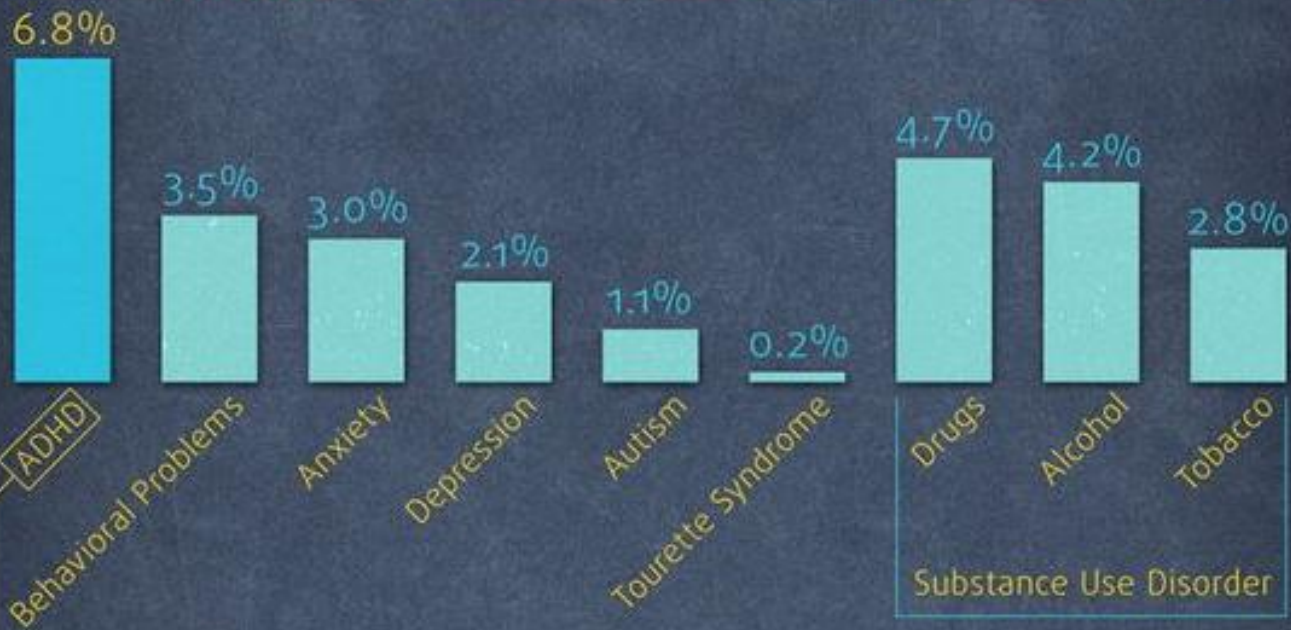


Suicide

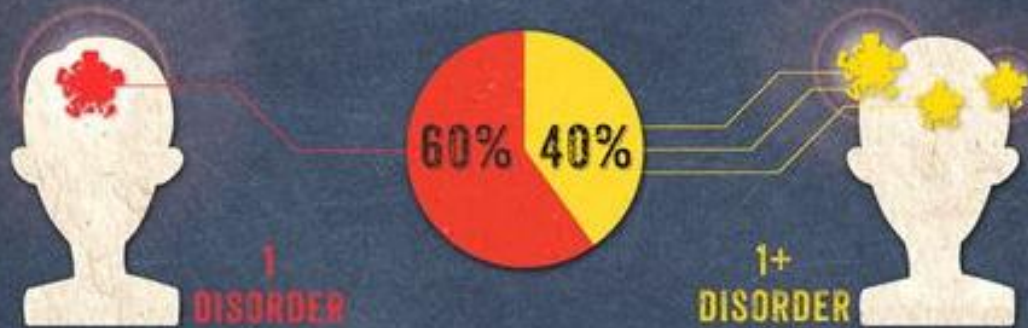
The second leading cause of death
among children aged 12-17 years.



PERCENT OF CHILDREN WHO CURRENTLY HAVE A MENTAL DISORDER



40% OF THOSE WITH A MENTAL DISORDER HAVE MORE THAN 1 DISORDER.



IT'S ONE OF THE COSTLIEST CONDITIONS TO TREAT IN CHILDREN.



\$247 BILLION

spent in healthcare cost every year for those with mental disorders under the age of 24.

▶ **ATTENTION DEFICIT HYPERACTIVITY DISORDER (ADHD)
IS THE LEADING MENTAL DISORDER AMONG CHILDREN.**

9% OF SCHOOL CHILDREN
HAVE ADHD COMPARED TO
5% OF THE POPULATION

OVERLOOKED CHILDREN

ADHD is oftentimes mistaken for behavioral problems.

50%

OF CHILDREN
WITH
ADHD
ARE ALSO...

**NEGATIVE
OR DEFIANT**

40%

OF CHILDREN
WITH
ADHD
ARE ALSO...

**DESTRUCTIVE
OR AGGRESSIVE**

25%

OF CHILDREN
WITH
ADHD
ARE ALSO...

**ANXIOUS
OR DEPRESSED**



OVERLOOKED FEMALES

ADHD affects both boys and girls at the same rate,
but girls get overlooked more often.

ATTENTION DEFICIT HYPERACTIVITY DISORDER

Girls usually exhibit
Inattention ADHD



Boys usually exhibit
Hyperactivity-Impulsivity ADHD



this
is more likely to get noticed and evaluated
than this



Boys are 3x more likely to be diagnosed than girls.



Men are 2x more likely to be diagnosed than women.

KNOW THE TRUTH

MYTH

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It can be difficult to separate the symptoms of a child's psychiatric disorder from a child's character. A psychiatric disorder is an illness not a personal flaw, just like diabetes or leukemia.

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THERAPY FOR KIDS IS A WASTE OF TIME.

Research has shown that treatment interventions have the best results during the first few years when symptoms of psychiatric disorders appear.

CHILDREN ARE OVERMEDICATED.

Skilled psychiatrists use great caution when deciding on a treatment plan that includes medication, which usually involves behavioral therapy. Medication is not the norm. Only 5% of children with psychiatric issues take medication.

CHILDREN GROW OUT OF MENTAL HEALTH PROBLEMS.

Children are less likely to "grow out" of psychiatric disorders than they are to "grow in" to more disability.

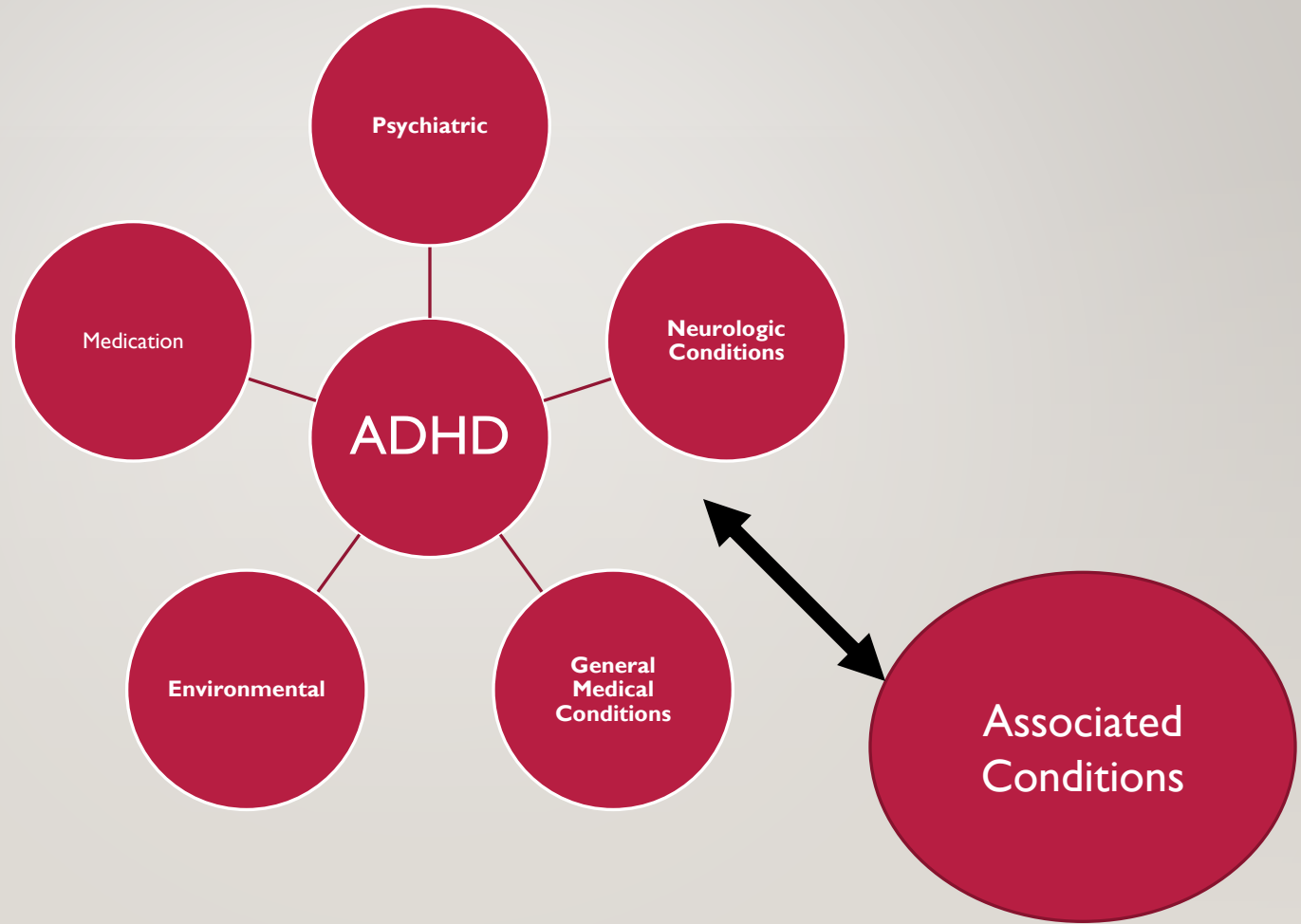
WHO IS THE PATIENT?

Greg



& the Family

Differential Diagnosis ADHD



Differential Diagnosis: Psychiatric Conditions



Differential Diagnosis: Psychiatric Conditions

- Low self esteem
- Major Depression
- Bipolar Disorder
- Dysthymia
- Anxiety Disorder
- Obsessive Compulsive Disorder
- Separation Anxiety
- Conduct Disorder
- Oppositional Defiant Disorder
- Poor Social skills
- Substance Abuse
 - 1. Consider if symptom onset in adolescence

General Medical Conditions



General Medical Conditions

- Hypothyroidism
- Severe Anemia
- Lead Poisoning
- Chronic illness
- Hearing Impairment or vision Impairment
- Obstructive Sleep Apnea in Children

Neurologic Conditions



Neurologic Conditions

Sleep disorders

- Obstructive Sleep Apnea
- Restless Leg Syndrome
- Periodic Limb Movement disorder

Tourette's Syndrome

Seizure disorder (e.g. Absence Seizure)

Sensory disorders

- Vision disorder
- Hearing Loss

Intellectual Disability

- Fragile X Syndrome
- Fetal Alcohol Syndrome
- Phenylketonuria
- Neurofibromatosis
- 22q11 Deletion Syndrome

Other learning disabilities

- Autism Spectrum Disorders
- Speech or language disorders



Environmental



Environmental

- Unsafe or disruptive learning environment
- School curriculum not well matched to child's ability
- Family dysfunction or poor parenting
- Child Abuse or neglect
- Parental Psychopathology

Medications



Medications

- Anticonvulsants
- Antihistamines
- Decongestants
- Beta agonists

Associated Conditions

- Language and Learning Disability (10-15%)
Usually less hyperactive and more inattentive
- Tourette's Syndrome(70% with tics have ADHD)
- Oppositional Defiant Disorder (35-65% of ADHD patients)
- Conduct Disorder (10-20% of ADHD patients)
- Major Depression (20% of ADHD patients)
- Anxiety Disorder (25% of ADHD patients)
- Substance Abuse (20-40% of ADHD patients)

WORK-UP: #1 IS A GOOD HISTORY

- **Consider other organic abnormality**
 - See ADHD Differential Diagnosis
 - Perinatal complications
 - Developmental Delay
 - Medical history
 - Family History of Attention Deficit Disorder
 - First degree relative with ADHD increases patient risk by 2-8 fold
- **Consider Learning Disability**
 - Ask educational history
 - **Is child reading at grade level?**
 - Is child's language at grade level?
 - School grades
 - Consider cognitive testing
- **Consider home environment**
 - Assess family stressors (e.g. financial, illness, single parent)
 - Assess family coping mechanisms

WORK-UP: #1 IS A GOOD HISTORY

-
- **Behavioral disorders**
 - School Truancy or absenteeism
 - Legal problems
 - **Consider comorbid psychiatric condition (comorbidity present in up to one third of ADHD patients)**
 - Major Depression or Anxiety Disorder
 - Excessive crying or worry
 - Preoccupation with death or Suicide
 - Conduct Disorder
 - Cruelty toward people or animals
 - Destruction of property or stealing
 - Delinquent behavior
 - Oppositional Defiant Disorder
 - Frequent rule breaking or arguments with adults
 - Frequently loses temper
 - Tourette's Syndrome
 - Repetitive vocal or Motor Tics
 - Sexual abuse history
 - Inappropriate sexual behavior
 - **Other contributing conditions**
 - See ADHD Comorbid Conditions
 - Sleep problems (e.g. Obstructive Sleep Apnea, Insomnia)

WORK-UP LAB

- CBC w/diff
- Comprehensive chemistry (fasting)
- Thyroid profile (TSH, free T4)
- Urine Drug Screen
- Pregnancy test
- EKG

LAB RESULTS

LAB RESULTS

- Check to see that the lab results are for the patient we are talking about.
- Different laboratories can have different normal ranges.
- If lab is ABNL consider redoing.
- If urine drug screen comes back not in your clinical suspicion,
redo with lab notified to monitor.
- ALWAYS initial the hard copy with date and time.

WORK-UP: SCALES

SCALES

- **Importance-**
diagnosis,
hypothesis-testing,
formulation,
assessment of patient's treatment progress
accurate assessment of symptoms

Common Behavior Rating Scales Used in the Assessment of ADHD and Monitoring of Treatment

Name of Scale	Reference
Academic Performance Rating Scale (APRS)	The APRS is a 19-item scale for determining a child's academic productivity and accuracy in grades 1–6 that has 6 scale points; construct, concurrent, and discriminant validity data as well as norms ($n = 247$) available (Barkley, 1990).
ADHD Rating Scale-IV	The ADHD Rating Scale-IV is an 18-item scale using <i>DSM-IV</i> criteria (DuPaul et al., 1998).
Brown ADD Rating Scales for Children, Adolescents, and Adults	Psychological Corporation, San Antonio, TX (www.drthomasebrown.com/assess_tools/index.html) (Brown, 2001)
Child Behavior Checklist (CBCL)	Parent-completed CBCL and Teacher-completed Teacher Report Form (TRF) www.aseba.org/index.html
Conners Parent Rating Scale–Revised (CPRS-R) ^a	Parent and adolescent self-report versions available (Conners, 1997)
Conners Teacher Rating Scale–Revised (CTRS-R) ^a	Conners, 1997
Conners Wells Adolescent Self-Report Scale	Conners and Wells, 1997
Home Situations Questionnaire–Revised (HSQ-R), School Situations Questionnaire–Revised (SSQ-R)	The HSQ-R is a 14-item scale designed to assess specific problems with attention and concentration across a variety of home and public situations; it uses a 0–9 scale and has test-retest, internal consistency, construct validity, discriminant validity, concurrent validity, and norms ($n = 581$) available (Barkley, 1990).
Inattention/Overactivity With Aggression (IOWA) Conners Teacher Rating Scale	The IOWA Conners is a 10-item scale developed to separate the inattention and overactivity ratings from oppositional defiance (Loney and Milich, 1982)
Swanson, Nolan, and Pelham (SNAP-IV) and SKAMP Internet site ADHD.NET	The SNAP-IV (Swanson, 1992) is a 26-item scale that contains <i>DSM-IV</i> criteria for ADHD and screens for other <i>DSM</i> diagnoses; the SKAMP (Wigal et al., 1998) is a 10-item scale that measures impairment of functioning at home and at school.
Vanderbilt ADHD Diagnostic Parent and Teacher Scales	Teachers rate 35 symptoms and 8 performance items measuring ADHD symptoms and common comorbid conditions (Wolraich et al., 2003a). The parent version contains all 18 ADHD symptoms, with items assessing comorbid conditions and performance (Wolraich et al., 2003b).

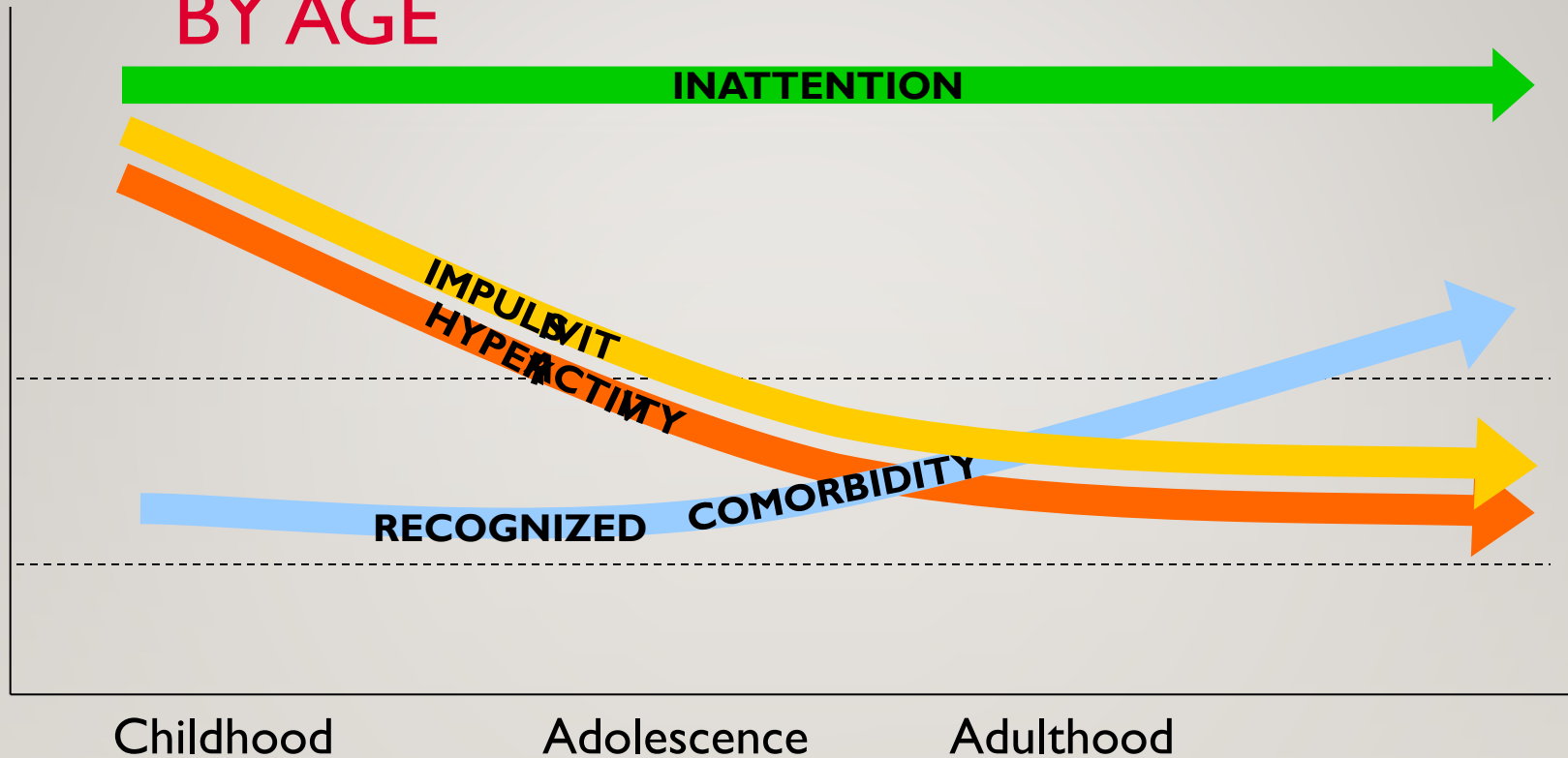
SCALES

Have teacher and parents fill out scales. You are looking for convergency of data.

Adult ADHD Self-Report Scale (ASRS-v1.1) Symptom Checklist

Patient Name		Today's Date					
Please answer the questions below, rating yourself on each of the criteria shown using the scale on the right side of the page. As you answer each question, place an X in the box that best describes how you have felt and conducted yourself over the past 6 months. Please give this completed checklist to your healthcare professional to discuss during today's appointment.			Never	Rarely	Sometimes	Often	Very Often
1. How often do you have trouble wrapping up the final details of a project, once the challenging parts have been done?							
2. How often do you have difficulty getting things in order when you have to do a task that requires organization?							
3. How often do you have problems remembering appointments or obligations?							
4. When you have a task that requires a lot of thought, how often do you avoid or delay getting started?							
5. How often do you fidget or squirm with your hands or feet when you have to sit down for a long time?							
6. How often do you feel overly active and compelled to do things, like you were driven by a motor?							
Part A							
7. How often do you make careless mistakes when you have to work on a boring or difficult project?							
8. How often do you have difficulty keeping your attention when you are doing boring or repetitive work?							
9. How often do you have difficulty concentrating on what people say to you, even when they are speaking to you directly?							
10. How often do you misplace or have difficulty finding things at home or at work?							
11. How often are you distracted by activity or noise around you?							
12. How often do you leave your seat in meetings or other situations in which you are expected to remain seated?							
13. How often do you feel restless or fidgety?							
14. How often do you have difficulty unwinding and relaxing when you have time to yourself?							
15. How often do you find yourself talking too much when you are in social situations?							
16. When you're in a conversation, how often do you find yourself finishing the sentences of the people you are talking to, before they can finish them themselves?							
17. How often do you have difficulty waiting your turn in situations when turn taking is required?							
18. How often do you interrupt others when they are busy?							
Part B							

ADHD DEVELOPMENTAL TRENDS BY AGE



TWO TYPES OF ADHD MEDICATIONS

Stimulants

Main target: Dopamine, a neurotransmitter that plays a key role in motivation. It also helps control movement and emotional responses.

Examples: Methylphenidates (like Concerta, Focalin, Metadate and Ritalin) and amphetamines (like Adderall and Vyvanse).

Usage: Stimulants are the most commonly prescribed type of ADHD medication.

Non-Stimulants

Main target: Norepinephrine, a neurotransmitter that plays a key role in executive functions like impulse control and getting started on tasks.

Examples: Atomoxetine (Strattera) and guanfacine (whose mode of action is not yet fully understood but appears to have an indirect effect on norepinephrine).

Usage: Non-stimulants tend to be used if kids don't respond well to stimulants. They can be used with or instead of stimulants.

CYP2D6 Pharmacokinetic Drug Interactions

Some ADHD medications are metabolized by the CYP2D6 enzyme. Concomitant use of these medications with substances known to inhibit CYP2D6 enzyme activity may result in increased levels of the ADHD medication.

ADHD Medications Metabolized by the CYP2D6 Enzyme

amphetamine salts (Adderall®)
dextroamphetamine (Dexedrine®)

lisdexamfetamine (Vyvanse®)
atomoxetine (Strattera®)

clonidine (Kapvay®)

Known Inhibitors of CYP2D6 Enzyme Activity

Concomitant use may increase the level of ADHD medications metabolized by the CYP2D6 enzyme

Antianginal

nicardipine
ranolazine

Antiarrhythmic

amiodarone
quinidine

Antibacterial

isoniazid

Anticholinergic

darifenacin

Antidepressant

bupropion
clomipramine
desipramine
duloxetine
fluoxetine
imipramine
paroxetine
sertraline

Antifungal

ketoconazole
miconazole
terbinafine

Antihistamine

diphenhydramine

Antimalarial

pyrimethamine
quinine

Antineoplastic

imatinib

Antiplatelet

ticlopidine

Antipsychotic

chlorpromazine
clozapine
haloperidol
thioridazine

Antiretroviral

delavirdine
ritonavir

Antithyroid

methimazole

Antiulcer

cimetidine

Hyperparathyroid

cinacalcet

Local Anesthetic

lidocaine

Psychostimulant

cocaine

Sedative

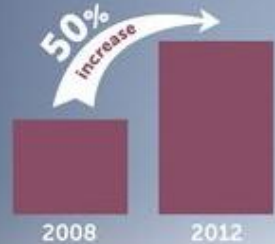
dexmedetomidine

HOW WIDESPREAD IS ADHD?



10 MILLION U.S. ADULTS ARE AFFECTED

NUMBER OF ADULTS TAKING ADHD MEDICATIONS



30% of children diagnosed with ADHD carry the disorder into adulthood

Yet in adults 26 and older, more women are medicated than men:

In children with ADHD, boys are medicated twice as often as girls:



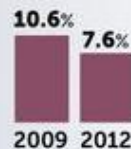
85% increase in adult ADHD prescriptions for women aged 26+ from 2009 to 2012

REASONS FOR THE RISE



Just 25% of adult ADHD patients were diagnosed in childhood

Use of antipsychotic medications in ADHD patients is decreasing:



PARENTS OF CHILDREN WITH ADHD ARE OFTEN DIAGNOSED AFTER RECOGNIZING THEY HAVE THE SAME SYMPTOMS AS THEIR CHILDREN



OVERLOOKED ADULTS



1/2-2/3



OF CHILDREN WITH ADHD
WILL STILL EXHIBIT SYMPTOMS AS ADULTS.

50%

OF DOCTORS SAY

I DO NOT FEEL CONFIDENT
DIAGNOSING ADHD IN ADULTS.



WHY IT'S HARD TO DIAGNOSE ADHD IN ADULTS

Some
presume it's a
character flaw.



Some
presume it's
normal.



Some
presume it's a
childhood/boy
thing.



Everyone
exhibits some
symptoms
sometimes.

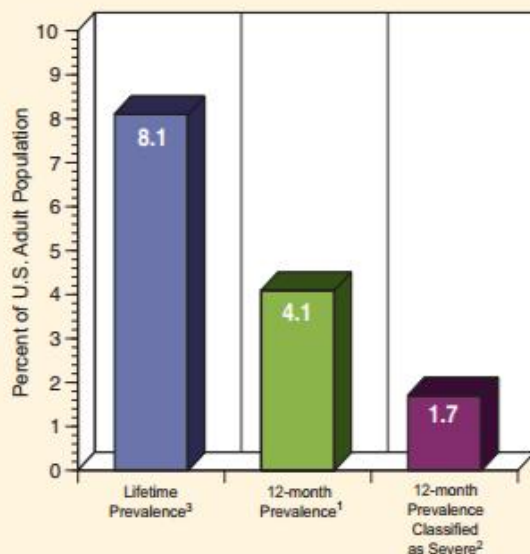
Have you ever been?

- ☒ Inattentive
- ☒ Hyperactive

Attention Deficit/Hyperactivity Disorder (ADHD)

Prevalence

- **12-month Prevalence:** 4.1% of U.S. adult population¹
- **Severe:** 41.3% of these cases (e.g., 1.7% of U.S. adult population) are classified as "severe"²



Demographics (for lifetime prevalence)⁵

- **Sex:** Not Reported
- **Race:** Not Reported
- **Age:**
 - 18–29: 7.8%
 - 30–44: 8.3%
 - 45–59: Not Reported
 - 60+: Not Reported

Average Age-of-Onset: 7 years old⁴

Treatment/Services Use

- **12-month Healthcare Use:** Not Reported
- **Percent Received Minimally Adequate Treatment:** Not Reported
- **12-month Any Service Use (including Healthcare):** Not Reported
- **Percent Received Minimally Adequate Treatment:** Not Reported

¹Kessler RC, Chiu WT, Demler O, Walters EE. Prevalence, severity, and comorbidity of twelve-month DSM-IV disorders in the National Comorbidity Survey Replication (NCS-R). *Archives of General Psychiatry*. 2005 Jun;62(6):617–27.

²Ibid

³Kessler RC, Berglund PA, Demler O, Jin R, Walters EE. Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the National Comorbidity Survey Replication (NCS-R). *Archives of General Psychiatry*. 2005 Jun;62(6):593–602.

⁴Ibid

⁵Ibid

The Most Prominent Learning Disabilities

Five learning disabilities are the most prominent among school-age children. Let's take a look at those five: their symptoms, possible causes and common issues.



Dyscalculia

What:

General mathematical disability in which person cannot comprehend quantity, positive and negative number values, fractions and many simple mathematical operations

Signs:

Issues with word problems, handling money, figuring out days/months, long division and patterns.



Dysgraphia

What:

Writing difficulties; usually linked to brain trauma

Signs:

Writing in a distorted way, consistently incorrectly spelling words and oddly spacing letters



Dyspraxia

What:

Motor skills problems

Signs:

Poor sense of balance and hand-eye coordination; clumsiness, frequently stumbling; irritation from loud noises and touch

In the brain:

Two hemispheres in the cerebral cortex do not share information or respond to each other properly as in an unaffected brain

FRONTAL LOBE

PARIETAL LOBE

TEMPORAL LOBE

ANGULAR GYRUS

OCCIPITAL LOBE



Dyslexia

What:

Reading disorder; recently linked to genetics

Signs:

Reading at lower levels than expected; inability to phonologically process words

In the brain:

Most likely angular gyrus, an area toward the back of the brain that understands and translates language



ADHD

What:

Inability to focus; most common learning disability

Signs:

Increased, random hyperactivity, impulsivity, short attention spans and disruptive behavior

In the brain:

Neurological, but doctors have yet to uncover the specific root of the issue inside the brain



Gerstmann's Syndrome

What:

Often seen after stroke or brain injury, this disorder can manifest itself via several different learning disabilities, including dysgraphia and dyscalculia

Signs:

Inability to differentiate between right and left; difficulty expressing thoughts in speech and writing

In the brain:

Left parietal lobe of the brain is damaged, in the angular gyrus region

NEURODEVELOPMENT DISORDER (INTELLECTUAL DISABILITY)



Learning Objectives:

After this presentation, you should be able to answer these questions:

- What is Intellectual Disability versus Global Developmental Delay?
- What is adaptive functioning, and why is it critical to understanding and helping children with Intellectual and Developmental Disabilities?
- How common are Intellectual and Developmental Disabilities? How does their prevalence vary by age, gender, and socioeconomic status (SES)?
- What are some known causes of Intellectual Disability in children?
- Why is studying children with specific behavioral phenotypes important?
- How might Intellectual and Developmental Disabilities be prevented?
- What educational interventions are available for children with Intellectual and Developmental Disabilities?
- How can behavioral and pharmaceutical treatments be used to reduce challenging behaviors in children in Intellectual Disability?
- What ethical issues are important to consider when selecting and implementing treatment?



CASE STUDY

DONTRELL: A FRIENDLY BOY

Dontrell was a 5-year-old African American boy referred to our clinic by his pediatrician. Dontrell showed delays in understanding language, speaking, and performing daily tasks. His mother had used alcohol and other drugs during pregnancy. She did not receive prenatal care because she was afraid that an obstetrician would report her drug use to the police. Dontrell was born with various drugs in his system and had respiratory and cardiovascular problems at birth. Shortly after delivery, Dontrell's mother disappeared, leaving him in his grandmother's care.

Dontrell was slow to reach many developmental milestones. Whereas most children learn to sit up by age 6 months and walk by their first birthday, Dontrell showed delays mastering each of these developmental tasks. Most striking were Dontrell's marked delays in language. Although he could understand and obey simple commands, he was able to speak only 15 to 20 words, and many of these were difficult to understand. He could not identify colors, was unable to recite the alphabet, and could not count. He also had problems performing self-care tasks typical of children his age. For example, he could not dress himself, wash his face, brush his teeth, or eat with utensils.

Dontrell showed significant problems with his behavior. First, he was hyperactive and inattentive. Second, Dontrell showed serious problems with defiance and aggression. When he did not get his way, he would tantrum and throw objects. He would also hit, kick, and bite other children and adults when he became upset. Third, Dontrell's grandmother said that he had "an obsession for food." Dontrell apparently had an insatiable appetite and was even caught hoarding food under his bed and stealing food from relatives.

Dr. Valencia, the psychologist who performed the evaluation, was most struck by Dontrell's appearance. Although only 5 years old, Dontrell weighed almost 85 lbs. He approached Dr. Valencia with a scowl and icy stare. Dr. Valencia extended her hand and said, "Hello." Dontrell grabbed Dr. Valencia's hand and kissed it! His grandmother quickly apologized, responding, "Sorry . . . he does that sometimes. He's showing that he likes you."



DIAGNOSTIC CRITERIA FOR INTELLECTUAL DISABILITY (INTELLECTUAL DEVELOPMENTAL DISORDER)

Intellectual disability (intellectual developmental disorder) is a disorder with onset during the developmental period that includes both intellectual and adaptive functioning deficits in conceptual, social, and practical domains. The following three criteria must be met:

- A. Deficits in intellectual functions, such as reasoning, problem solving, planning, abstract thinking, judgment, academic learning, and learning from experience, confirmed by both clinical assessment and individualized, standardized intelligence testing.
- B. Deficits in adaptive functioning that result in failure to meet developmental and socio-cultural standards for personal independence and social responsibility. Without ongoing support, the adaptive deficits limit functioning in one or more activities of daily life, such as communication, social participation, and independent living, across multiple environments, such as home, work, and community.
- C. Onset of intellectual and adaptive deficits during the developmental period.

Specify current severity: Mild, Moderate, Severe, Profound*



There has been considerable controversy regarding the name of the disorder “Intellectual Disability.” In DSM-IV, this disorder was called “Mental Retardation” to reflect the below-average intellectual ability of individuals with this condition. However, the developers of DSM-5 agreed to abandon this term because of its negative connotation. When revising DSM-IV, the American Psychiatric Association’s Neurodevelopmental Disorders working group considered renaming the disorder “Intellectual Developmental Disorder.” Their proposal drew considerable criticism from the leading professional organization of individuals who work with people with developmental disabilities, the American Association of Intellectual and Developmental Disabilities (AAIDD; Gomez & Nygren, 2012). Instead, The AAIDD argued that the name “Intellectual Disability” (not Developmental) be adopted in DSM-5 for several reasons:

- Intellectual Disability is the most used term in the United States and internationally to refer to people with intellectual and adaptive skills deficits.
- The term reflects the World Health Organization’s conceptualization of low intelligence and adaptive functioning as a “disability.”
- It implies deficits in both intelligence and adaptive functioning, not only low IQ.
- It is less offensive than the often-pejorative term Mental Retardation.

In 2010, Rosa’s Law (PL 111-256) replaced the term Mental Retardation with Intellectual Disability in federal education, health, and labor laws. The law was named after 9-year-old Rosa Marcellino, a girl with Down Syndrome, whose family worked to have the term retardation removed from the educational code in her home state of Maryland.

The APA decided to adopt the term Intellectual Disability in DSM-5 yet retains the term Intellectual Developmental Disorder in parentheses.



ADAPTIVE FUNCTIONING EXAMPLES

Intellectual Disability. Adaptive functioning scales allow clinicians to assess children's conceptual, social, and practical skills. Caregivers' reports are converted to standard scores which can be used to determine if children have deficits compared to other children their age in the general population. Below are some areas of adaptive functioning that might be assessed in younger children, older children, and adolescents

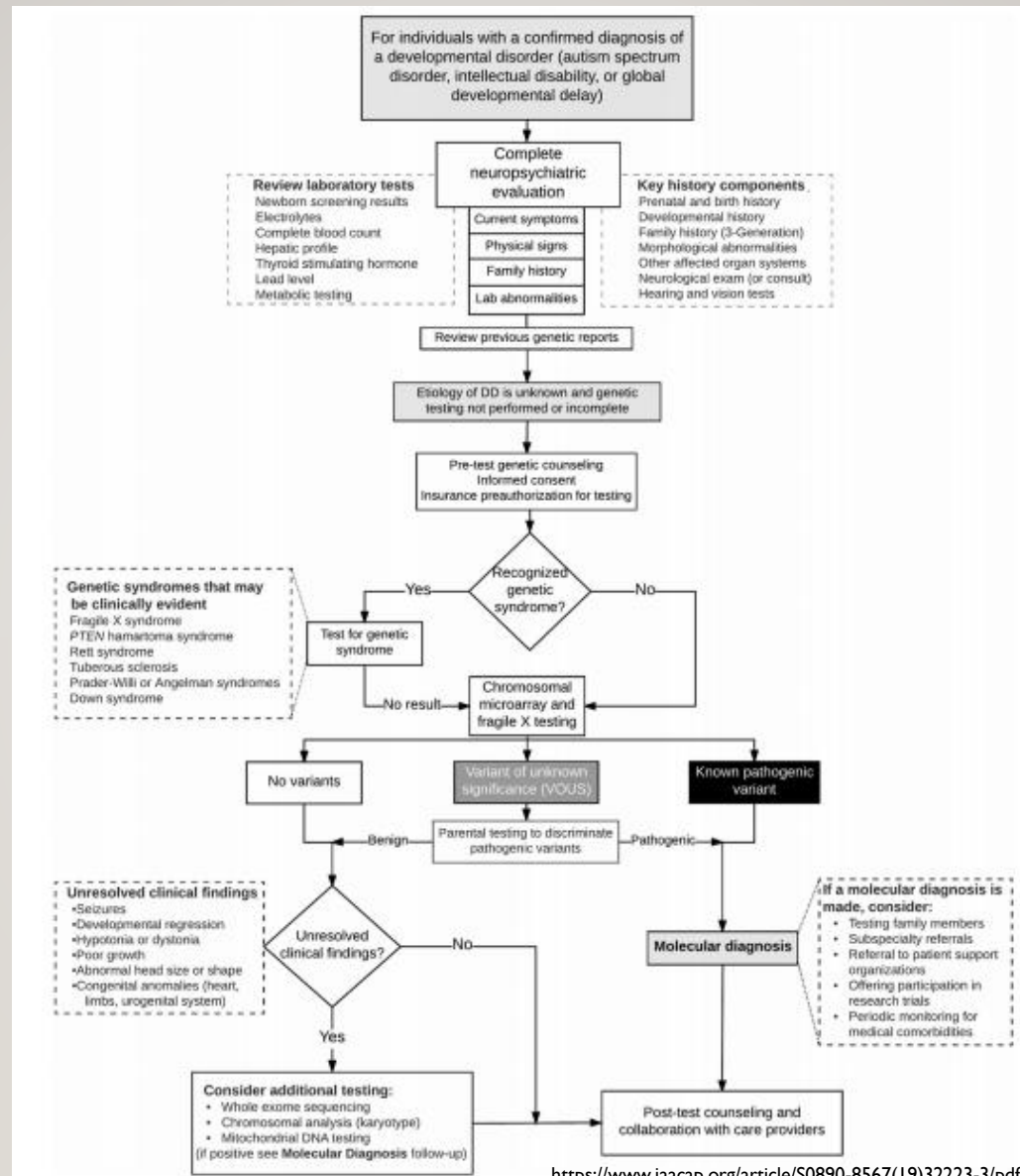
	Younger Children	Older Children	Adolescents
Conceptual	Can count 10 objects, one by one; Knows day, month, year of birth	States value of penny, nickel, dime; Uses mathematical operations	Sets a watch or clock to correct time; Can complete a job application
Social	Says "hi" and "bye" when coming and going; Asks for help when needed	Reads and obeys common signs (e.g., stop, do not enter); Knows topic of group conversations	Has satisfying friendships; Keeps personal information private
Practical	Uses the restroom; Drinks from a cup without spilling	Answers the telephone; Can safely cross busy street	Travels to school or work by themselves; Washes clothes, dishes

Describing the Severity of Intellectual Disability

Severity	Conceptual Domain	Social Domain	Practical Domain
Mild	Preschoolers may show no obvious conceptual differences. School-aged children show difficulties in acquiring academic skills (e.g., reading, writing, arithmetic, telling time, using money). Abstract thinking and planning may be impaired; thinking tends to be concrete.	Communication, conversation, and language are more concrete or immature than the skills of peers. The child may have difficulty accurately understanding the social cues of others. There may be difficulties regulating emotion and behavior compared to peers.	The child may function in an age-expected manner with regard to personal care. In adolescence, assistance may be needed to perform more complex daily living tasks like shopping, cooking, and managing money.
Moderate	Preschoolers' language and preacademic skills develop slowly. School-age children show slow progress in academic skills. Academic skill development is usually at the elementary school level.	The child shows marked differences in social and communicative skills compared to peers. Spoken language is simplistic and concrete. Social judgment and decision making are limited. Friendships with peers are often affected by social or communicative deficits.	The child needs more time and practice learning self-care skills, such as eating, dressing, toileting, and hygiene, than peers. Household skills can be acquired by adolescent with ample practice.
Severe	The child generally has little understanding of written language or numbers. Caretakers must provide extensive support for problem solving throughout life.	There are limited spoken language skills with simplistic vocabulary and grammar. Speech may be single words/phrases. The child understands simple speech and gestures. Relationships are with family members and other familiar people.	The child needs ongoing support for all activities of daily living: eating, dressing, bathing, elimination. Caregivers must supervise at all times. Some youths show challenging behaviors, such as self-injury.
Profound	Conceptual skills generally involve the physical world rather than symbols (e.g., letters, numbers). Some visuospatial skills, such as matching and sorting, may be acquired with practice. Co-occurring physical problems may greatly limit functioning.	The child has limited understanding of symbolic communication. The child may understand some simple instructions and gestures. Communication is usually through nonverbal, non-symbolic means. Relationships are usually with family members and other familiar people. Co-occurring physical problems may greatly limit functioning.	The child is dependent on others for all aspects of physical care, health, and safety, although he or she may participate in some aspects of selfcare. Some youths show challenging behaviors, such as self-injury. Co-occurring physical problems may greatly limit functioning.

Source: DSM-5, 2013.

Diagnostic Genetic Testing Algorithm for Youth With Developmental Disorders



What Is Intellectual Disability?

- Intellectual Disability is characterized by significant limitations in cognitive and adaptive functioning that emerge during the course of children's development.
 - IQ scores <70–75 usually indicate limitations in cognitive ability.
 - Adaptive functioning is measured in three broad domains (a) conceptual skills, (b) social skills, and (c) practical skills. Scores <70–75 usually indicate limitations in adaptive functioning.
 - Cognitive and adaptive limitations are equally important to the identification of Intellectual Disability. Children are not diagnosed based on IQ alone.
- The American Association on Intellectual and Developmental Disabilities (AAIDD) offers its own definition on Intellectual Disability. It is largely consistent with the *DSM-5* definition, but it places greater emphasis on contextual factors and the supports children need to succeed academically and socially.
- Global Developmental Delay is characterized by significant deficits in two of the following domains that emerge prior to age 5 years: (a) fine/gross motor skills, (b) speech/language, (c) social/personal skills, and (d) daily living skills.
 - Global Developmental Delay is usually diagnosed in infants and very young children suspected of Intellectual Disability but who are too young to be administered an IQ test.
 - Not all children with Global Developmental Delay will meet criteria for Intellectual Disability when they become older.

Which of the following is true of children with global developmental delay (GDD)?

- A. All will go on to be diagnosed with intellectual disability.
- B. It is called a delay because those with GDD will eventually catch up to their peers.
- C. They are usually diagnosed within the first year of life.
- D. They almost always have a genetic disorder such as Down syndrome that explains their problem.

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Associated Characteristics

- Many youths with Intellectual and Developmental Disabilities show challenging behavior: actions of sufficient intensity, frequency, or duration that children's physical safety (or the safety of others) is jeopardized. Challenging behaviors include the following:
 - Stereotypies are actions performed in a consistent, rigid, and repetitive manner with no immediate practical significance (e.g., hand flapping, rocking back and forth).
 - Self-injurious behaviors are repetitive actions that can, or do, cause physical harm to the person.
 - Physical aggression are actions that cause (or can cause) property destruction or injury to others.
- Children with Intellectual and Developmental Disabilities can experience other psychiatric disorders and are at greater risk than typically developing children their age.

Epidemiology

- Approximately 1.8% of individuals have Intellectual Disability. The disorder is more common among school-age children than adults and is more common among boys than girls.
- According to Edward Zigler, children with Intellectual Disability can be differentiated into two types:
 - Youths with organic Intellectual Disability have known medical causes for their impairments. They tend to have IQ scores <50, physical abnormalities associated with neurological problems, and parents without impairments.
 - Youths with cultural-familial Intellectual Disability have no identified cause for their impairments. They tend to have IQ scores between 50 and 70, have normal physical appearance, have parents with low intelligence, and come from lower income families.
- According to the similar sequence hypothesis, children with Intellectual Disability show cognitive development in the same sequence as typically developing children, albeit at a slower pace. Most research supports this hypothesis.
- According to the similar structure hypothesis, children with and without Intellectual Disabilities who have the same mental age will show the same level of abilities. Although generally supported, some children with organic Intellectual Disability show abilities different than their typically developing peers.
- A behavioral phenotype refers to the curious pattern of strengths and weaknesses shown by children with specific causes for their Intellectual Disability.

Causes of Intellectual Disability

- Chromosomal abnormalities can cause Intellectual Disability.
 - Down Syndrome is usually caused by an extra 21st chromosome (trisomy 21). It is characterized by Intellectual Disability, relative deficits in language, and relative strengths in visual-spatial skills and social functioning in childhood.
 - Fragile X Syndrome is caused by an X-link genetic mutation. The disorder is more severe in boys. It is characterized by Intellectual Disability, relative deficits in sequential processing, relative strengths in simultaneous processing, and problems with social functioning.
 - Prader-Willi Syndrome is caused by a missing paternal chromosome 15. It is characterized by mild Intellectual Impairment, obesity, and obsessive-compulsive behavior.
 - Angelman's Syndrome is caused by a missing maternal chromosome 15. It is characterized by Intellectual Disability, relative deficits in language, and unusually happy demeanor.
 - William's syndrome is caused by deletions on chromosome 7. It is characterized by Intellectual Disability, relative weakness on visual-spatial tasks, well-developed vocabularies and spoken language, and inattention/hyperactivity.
- Metabolic disorders can also cause Intellectual Disability. PKU is caused by a recessive gene which can cause severe disability if children do not abstain from phenylalanine-rich foods (e.g., cheese, meats).
- Teratogen exposure in utero can cause Intellectual Disability. Common teratogens include bacteria, viruses, alcohol, and other drugs.
- Complications during pregnancy or delivery can contribute to Intellectual Disability, especially when they restrict the fetus's access to oxygen.
- Childhood illnesses or injuries can cause Intellectual Disability including encephalitis, meningitis, high fever, and lead toxicity.
- Cultural-familial Intellectual Disability is usually caused by a genetic propensity toward below-average cognitive functioning and environmental deprivation or socioeconomic disadvantage.

Prevention and Treatment

- Women at high risk may participate in prenatal screening to estimate the likelihood of having a child with a genetic disorder. Screening techniques include (a) serum screening, (b) amniocentesis, and (c) chorionic villus sampling.
- Prevention programs for youths at risk for Intellectual Disability, such as the Infant Health and Development Program, are associated with short-term gains in IQ, but, no long-term differences in children's outcomes.
- The Individuals with Disabilities Education Improvement Act requires children with Intellectual Disability to be identified at an early age and provided with services to help them achieve.
 - School-age children receive an Individualized Education Program (IEP) and usually participate in academic inclusion programs alongside typically developing peers.
 - Universal design is an educational practice that involves instructional materials and activities that allow learning goals to be achievable by children with and without disabilities.
- Applied Behavior Analysis (ABA) is typically used to reduce challenging behavior in children with Intellectual Disability.
 - Clinicians can use event recording, interval recording, or duration recording to identify and monitor challenging behavior.
 - Functional analysis can help determine the cause of the behavior:
 - Positive social reinforcement (i.e., to get attention)
 - Negative reinforcement (i.e., to avoid or escape unpleasant tasks)
 - Automatic reinforcement (i.e., self-stimulation)
 - Clinicians can use operant conditioning to correct behavior.
 - Positive reinforcement (e.g., differential reinforcement)
 - Positive punishment (e.g., overcorrection, positive practice)
 - Negative punishment (e.g., time out, response cost)
- Medication, such as risperidone (Risperdal) can be used to reduce some challenging behavior.
- Therapists can provide information to families of children with Intellectual and Developmental Disabilities, alleviate parenting stress, and teach problem-focused coping.

System	Term	Definition	Intellectual functioning	Adaptive behavior	Developmental period
ICD-10 ⁴	Mental retardation	It is a condition of arrested or incomplete development of the mind, which is especially characterized by impairment of skills manifested during the developmental period, which contribute to the overall level of intelligence, i.e., cognitive, language, motor, and social abilities	Components are cognition, language, and motor and social skills An intelligence quotient of 70 is the cutoff It categorizes ID into four severity levels that are based on IQ	Not clearly defined It is implied that assessment of adaptive behavior is part of assessment of intellectual functioning	Not explicitly defined, but understood to consider it as 18 years
DSM-5 ⁵	Intellectual disability (intellectual developmental disorder)	Intellectual disability (intellectual developmental disorder) is a disorder with onset during the developmental period that includes intellectual and adaptive functioning deficits in conceptual, social, and practical domains	Components are reasoning, problem-solving, planning, abstract thinking, judgment, academic learning, and learning from experience On standardized tests of intelligence, score of 65-75 is considered to indicate intellectual disability, where the test quotients have standard deviation of 15 and mean of 100 and standard error of 5	Deficits on adaptive functioning result in failure to meet developmental and sociocultural standards for personal independence and social responsibility Without ongoing support, the deficits will affect one or more activities of daily life, such as communication, social participation, and independent living across multiple environments Nomenclature of severity levels is same as it is in ICD-10, but the levels are decided based on the deficits only in adaptive functioning DSM-5 has explained the adaptive behaviors in each of the three domains of intellectual functioning such as conceptual, social, and practical domains in reference to the severity level and age	Defined as 18 years
Rights of Persons with Disabilities Act, 2016 ⁶	Intellectual disability	Intellectual disability, a condition characterized by significant limitation both in intellectual functioning (reasoning, learning, and problem-solving) and in adaptive behavior which covers a range of day-to-day, social, and practical skills	Like ICD-10, it has adopted the IQ cutoff of 70 for ID, and the same terminology to denote severity levels, but with different cutoffs. The severity levels are based on the scores of the Vineland Social Maturity Scale (a standardized, normative measure adaptive behavior scale) Profound disability=0-20 (100%) Severe=21-35 (90%) Moderate=36-54 (75%) Mild=55-69 (50%) Borderline=70-84 (25%) Note: Borderline disability is not a benchmark disability.	Adaptive behavior is not defined but is understood to cover a range of day-to-day, social, and practical skills Scores on Vineland Social Maturity scale (a standardized, normative measure adaptive behavior scale) are considered to define the severity of ID	Not explicitly defined, but understood to consider it as 18 years
ICD-11 Working Group on Intellectual Disability (2011) ^{4,7}	Intellectual developmental disorders	A group of developmental conditions characterized by significant impairment of cognitive functions, which are associated with limitations of learning, adaptive behavior, and skills	The working group advocated continuing clinical severity levels mentioned in ICD-10 due to their diagnostic and clinical utility. Therefore, IQ score should be considered as a clinical descriptor among others that are considered important in determining the severity levels of ID	Adaptive behavior is not defined but implied that difficulties in adaptive behavior will manifest in meeting the demands of daily life expected for one's age peers, cultural, and community environment. These difficulties include limitations in relevant conceptual, social, and practical skills	Not explicitly defined, but understood to consider it as 18 years

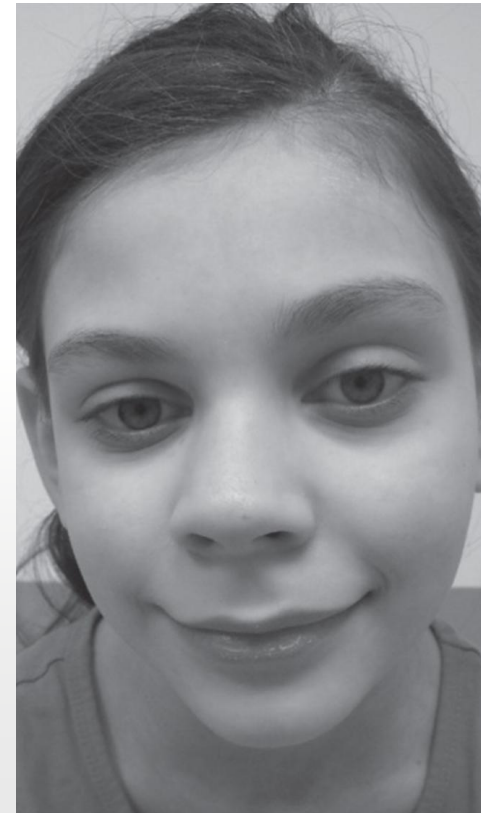
⁴World Health Organization (1992). ICD-10 Classification of Mental and Behavioural Disorders. Geneva: World Health Organization. ⁵American Psychiatric Association (2013). Diagnostic and Statistical Manual of Mental Disorders, 5th ed. Arlington, VA: American Psychiatric Association. ⁶Government of India. Rights of Persons with Disabilities Act. New Delhi: Government of India, 2016. ⁷Salvador Carulla L, Reed GM, Vaez-Aziz LM, Cooper S-A, Martinez-Leal R, Bertel M, et al. Intellectual developmental disorders: Towards a new name, definition and framework for "mental retardation/intellectual disability" in ICD-11. World Psychiatry 2011; 10:175-180. ⁸Need to refer to ICD-11 final guidelines as and when they become operational. ICD-10 – International Classification of Diseases, 10th revision; DSM-5 – Diagnostic and Statistical Manual of Mental Disorders, 5th Edition. ID – Intellectual disability. IQ – Intelligence quotient

Commonly Used Screening Tools for Developmental Delay

Tool	Age Range	Description	Psychometrics	Availability
Ages and Stages Questionnaire (ASQ-III)	1–66 mo	30-Item age-specific parent report measure Examines communication, gross motor, fine motor, problem solving, personal-social, self-regulation, compliance, language, adaptive behaviors, autonomy, affect, and interaction with people	High test–retest reliability (.92) and interrater reliability (0.93) for risk classification. Sensitivity 0.83–0.89 Specificity 0.80–0.92 across ages	Available for purchase at: http://products.brookespublishing.com/ASQ-3-Starter-Kit-P574.aspx
Parents' Evaluation of Developmental Status (PEDS)	1–95 mo	10-Item parent report measure. Examines global/cognitive, expressive language, receptive language, fine motor, gross motor, self-help, and social–emotional	Sensitivity 0.74–0.79 Specificity 0.70–0.80 across ages	Available for purchase at: http://www.pedstestshop.com/product-category/peds-products/
Bayley Infant Neurodevelopmental Screener (BINS)	3–24 mo	10- to 15-min screen for neurological impairment or developmental delay	Test–retest reliability 0.71–0.81 Strong internal consistency Interrater reliability 0.79–0.96	Available for purchase at: http://www.pearsonclinical.com/psychology/products/100000163/bayley-infant-neurodevelopmental-screener-bins-bins.html
Denver Prescreening Developmental Questionnaire (Denver PDQ-II)	2 wk to 6 y	Parent report measure. Positive results indicate follow-up testing with Denver-II by a professional. Examines personal and social, fine motor, gross motor, and language	Sensitivity 83% Specificity 43%	Available for free download at: http://denverii.com/pdq-iis-english-birth-to-6-yrs-instructions/
Infant/Toddler Checklist (ITC)	6–24 mo	Population-based screener designed to identify risk for language and communication impairment	Sensitivity 0.87–0.94; specificity 0.75–0.89	Available for free download at: http://www.brookespublishing.com/resource-center/screening-and-assessment/csbs/csbs-dp/csbs-dp-itc/

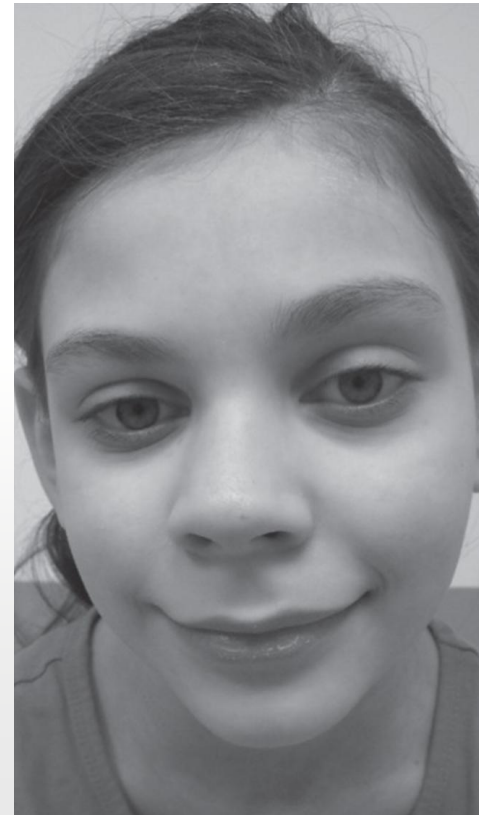
[https://www.jaacap.org/article/S0890-8567\(19\)32223-3/pdf](https://www.jaacap.org/article/S0890-8567(19)32223-3/pdf)

- An 11-year-old Caucasian girl was referred for evaluation of aggressive behavior that had worsened over the past two years.
 - She experienced frequent outbursts of anger and would scream or bite when she became upset.
 - She was cognitively delayed and could not independently perform activities of daily living such as bathing and eating.
 - Attempts at toilet training were unsuccessful.
 - She exhibited repetitive mannerisms such as hand flapping, eye blinking, repeated hand washing and running in circles.
 - Socialization skills were restricted, and eye contact was poor.
-
- Family history was significant for four brothers with developmental delay, a maternal aunt with premature menopause, and history of gait instability and tremors in her maternal grandfather since 40 years of age.
 - There was no history of consanguinity.
 - No complications were reported during pregnancy or birth.
 - Her motor milestones were normal.
 - Although she exhibited language delay, she could engage in a conversation with diminished vocabulary.



<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC427681/>

- On physical examination, vital signs were normal.
- Subtle facial dysmorphism was present
- Her head was normocephalic, and cardiac and pulmonary examinations were unremarkable.
- The patient was noted to be intellectually impaired, hyperactive, had poor eye contact and displayed repetitive behavior.
- The remainder of the neurological examination (cranial nerves, motor, sensory and reflexes) was normal.



<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC427638>

FRAGILE X SYNDROME (FXS)

- This case is a classic presentation of fragile X syndrome (FXS). FXS is caused by a mutation of the fragile X mental retardation 1 (*FMR1*) gene on Xq27.3. The vast majority of cases occur as a result of unstable expansion of the CGG repeat in the *FMR1* gene. The presence of >200 repeats is associated with hypermethylation, leading to transcriptional silencing and a decrease or absence of the gene product, fragile X mental retardation protein (1). The diagnosis of FXS is made by assessing the number of CGG repeats using polymerase chain reaction analysis, usually in conjunction with Southern blot analysis, which evaluates the methylation status of the gene. The patient in the present case was genetically confirmed to have the full mutation (667 repeats of CGG), along with four of her brothers.
- FXS is the most common cause of inherited intellectual disability and is strongly associated with autism (1). It may present with features of pervasive developmental disorder including language delay, stereotypic behaviors and social impairment. Children with FXS often experience scholastic problems due to autism in addition to the underlying intellectual disability, and are frequently placed in special education classes or are homeschooled. Other features include attention deficit hyperactivity disorder, hyperextensible joints, mitral valve prolapse, flat feet, hypersensitivity to sensory stimuli and hypotonia (1). Many patients with FXS, particularly boys, develop seizures in early childhood (1). Individuals with FXS often have a characteristic physical appearance including macrocephaly, a long face with a prominent forehead and large ears. Post pubertal males can develop macro-orchidism. Similar to our patient, females affected with FXS may exhibit subtle dysmorphic features. As a result of random X inactivation and presence of *FMR1* from the normal X chromosome, females with FXS may develop an attenuated phenotype (1). It may, thus, be challenging to suspect FXS in a girl solely based on clinical presentation.
- Family history may provide important clues to the diagnosis of FXS. While the presence of >200 CGG repeats leads to FXS, patients with 55 to 200 repeats ('premutation') have a different clinical phenotype. Premutation carrier males often have mild cognitive deficits and may develop fragile X-associated tremor/ataxia syndrome (FXTAS), which is a Parkinson's-like condition that presents with gait instability and postural tremors. Premutation carrier females may develop premature ovarian failure, anxiety and depression (2). The symptoms reported in the patient's maternal grandfather and aunt were suspicious for FXTAS and premature ovarian failure, respectively. Taking a thorough history is vital to obtain these important clues, which may assist in evaluation and diagnosis.
- Interventions, such as early developmental stimulation, physical therapy, occupational therapy, and speech and language therapy, are key to the management of children with FXS. Stimulants and anti-depressants can be prescribed for behavioral symptoms including hyperactivity, inattention and mood disturbances. Children with seizures should be treated with anticonvulsants. Targeted therapeutic options are currently being evaluated, including R-baclofen, which may have a role in decreasing seizures, improving autistic behaviors and decreasing social impairment (2). Families should be offered genetic counselling to determine the extent of the mutation and risk of transmission because there are implications for the diagnosis on seemingly unaffected family members.
- In summary, FXS remains the most common cause of inherited intellectual disability and should be a part of the evaluation of any child with unexplained developmental delay. Our patient exhibited some of the physical characteristics of FXS, but the presence of pervasive features along with the intellectual disability and a strong family history favored a diagnosis of FXS. It is important to consider FXS in girls presenting with intellectual disability because the clinical features may be more subtle compared with boys.

1. Penagarikano O, Mulle JG, Warren ST. The pathophysiology of Fragile X Syndrome. *Annu Rev Genomics Hum Genet.* 2007;8:109–29.
2. Berry-Kravis E, Knox A, Hervey C. Targeted treatments for fragile X syndrome. *J Neurodev Disord.* 2011;3:193–210.

CLINICAL PEARLS

- FXS is the most common inherited cause of intellectual disability.
- Characteristic features of FXS include physical characteristics such as a long face, large ears and macro-orchidism (in postpubertal males), as well as autism, seizures and attention deficit hyperactivity disorder. Premutation carriers can present with FXTAS (males) and premature ovarian failure (females).
- FXS should be considered as a potential diagnosis in girls presenting with unexplained intellectual disability because other features of the disease may be less conspicuous.

Fragile X syndrome is associated with which of the following?

- a) Language impairment
- b) Behavioral problems
- c) Moderate levels of intellectual disability
- d) All of the above

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- a) Language impairment
- b) Behavioral problems
- c) Moderate levels of intellectual disability
- d) All of the above



Ron is an 8-year-old boy brought to the office with parents saying he has jerky movements; speech was delayed until 3 years and cognitive disability and epilepsy. Parents report he is happy even when he shouldn't be. Your thoughts?





How do you work him up?





How do you work him up?

PE-

CBC w/diff
Comprehensive Chem
TSH

EEG

Brain MRI



How do you work him up?

PE- mandibular prognathism, strabismus and unusual laughing facial expression Head circ- 54 cm

CBC w/diff- nl

Comprehensive Chem- nl

TSH- elevated

Brain MRI- nl

EEG- Spike and Wave pattern

NOW WHAT?? What do you expect to find?



Genetic testing-

What do you expect to find? Which Chromosome?



Genetic testing-

What do you expect to find? Which Chromosome?

Chromosome 15- deletion



Ron has impaired communication, which might occur due to language deficit and intellectual dysfunction

Delayed motor development is common in AS, jerky movements are the first manifestation of AS, recognize by parents. Ron's parents report started at the age of 2 yr.

Facial abnormalities wide mouth, abnormal teeth, tongue protrusion and mandibular prognathism.

Around 70% of AS children could walk at 3 yr old. Ron walked walk independently at 4 yr, but he still cannot run at 8.

EEG showed generalized spike and wave pattern .Over 85% of AS patients develop seizure in the first 3 yr of life. Onset varies from 1 month to 20 years. Epilepsy is severe in AS cases and is hard to control as our case. Valproic acid is the therapeutic choice in AS epilepsy.

What is the relationship between the cause of Prader-Willi syndrome and the cause of Angelman syndrome?

- A. Both involve chromosome 15; Prader-Willi involves a deletion of genetic material on this chromosome, while Angelman Syndrome involves extra material on this chromosome.
- B. Both involve chromosome 15; Prader-Willi involves a deletion of genetic material from the father, whereas Angelman involves a deletion of genetic material from the mother.
- C. Both involve the deletion of genetic material; Prader-Willi from chromosome 15 and Angelman from chromosome 16.
- D. Both involve nondisjunction; Prader-Willi on chromosome 15 and Angelman on chromosome 16.



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Newborn

- Hypotonia

- Feeding problems and FTT

Infancy

- Hyperphagia

- distinctive facial features

Childhood and adolescence

- hypogonadism

- developmental delay

Adulthood

- eating disorder

Your working Diagnosis?



Newborn

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Prader Willi

Genetic testing

Genetic testing for Prader-Willi syndrome (PWS) includes chromosomal or microarray analysis and assessment for methylation patterns in the Prader-Willi syndrome region.

Methylation patterns can be determined with Southern blot hybridization or polymerase chain reaction (PCR) using DNA primers that can detect methylated cytosine.

Analysis for underlying uniparental disomy requires samples from both parents and the child with Prader-Willi syndrome.

Fluorescent in situ hybridization (FISH) can be used to confirm prenatal diagnosis when a deletion in the 15q region is suspected after chorionic villus sampling or amniocentesis.

In a patient with an imprinting center mutation, test both biological parents for the presence of asymptomatic mutations in the imprinting center; such mutations indicate a higher risk for recurrence.



Hypogonadism

Most patients with Prader-Willi syndrome have hypothalamic dysfunction that manifests as short stature, central obesity, hypogonadism, and osteoporosis.

Fasting measurements of serum insulinlike growth factor-1 (IGF-1) and insulinlike growth factor binding protein-3 (IGFBP-3) levels are good screening measurements for underlying growth hormone deficiency.

Refer patients with diminished growth velocity and abnormal levels of IGF-1 and IGFBP-3 to a pediatric endocrinologist for provocative growth hormone stimulation testing.

Assess thyroid and adrenal status in patients when clinically warranted.

Hypopituitarism has been reported in some patients with Prader-Willi syndrome.



Obesity

Measure glycosylated hemoglobin in patients with Prader-Willi syndrome who are obese to assess for the development of type 2 diabetes mellitus as clinically warranted, especially if the patient is taking growth hormone supplementation.

Evaluate patients with Prader-Willi syndrome for biochemical evidence of pickwickian syndrome (eg, hypercarbia, polycythemia) as clinically warranted.

If symptoms suggest sleep apnea or narcolepsy, perform a sleep study with multiple sleep latency testing.



Imaging Studies

Individuals with Prader-Willi syndrome are at risk for pathologic fractures secondary to underlying osteoporosis. A high pain tolerance may allow for minimal symptoms of discomfort with obvious deformity. Patients with Prader-Willi syndrome may require the following imaging studies:

MRI of the head (to evaluate for hypopituitarism)

Serial dual energy x-ray absorptiometry (DEXA) scanning (for detection and monitoring of osteoporosis)

Scoliosis films



Imaging Studies

Chest radiography (if cor pulmonale is suspected)

Other imaging modalities as clinically dictated (eg, extremity film for limp evaluation, hip films to screen for hip dysplasia)

In patients who suddenly develop abdominal distention, abdominal pain, or a decrease in appetite, imaging including plain abdominal radiography, abdominal ultrasonography, or CT scanning and gastrointestinal series may be warranted to screen for possible conditions such as acute gastric dilation, cholelithiasis, or pancreatitis.



Procedures

Assess the growth hormone axis and adrenal axis under the supervision of an endocrinologist if clinically warranted

Common Measures of Intellectual Functioning in Children and Adolescents

Instrument and current version	Age Range	Description:	Availability
Bayley Scales of Infant and Toddler Development (BSID-III)	2–49 mo	Subtests: language, receptive and expressive, gross motor, fine motor, cognitive problem-solving ability, and sustained attention	Available for purchase at: http://www.pearsonclinical.com/childhood/products/100000123/bayley-scales-of-infant-and-toddler-development-third-edition-bayley-iii.html
Stanford–Binet Intelligence Scale (SB5)	2–85 y	Gives an FSIQ. Subtests: verbal reasoning, abstract/visual reasoning, quantitative memory, and short-term memory	Available for purchase at: http://www.wpspublish.com/store/p/2951/stanford-binet-intelligence-scales-fifth-edition-sb-5
Wechsler Preschool and Primary Scale of Intelligence-Revised (WPPSI-R)	2 y 6 mo to 7 y 3 mo	Gives an FSIQ score. Subtests: verbal, processing, performance, and general language.	Available for purchase at: http://www.pearsonclinical.com/psychology/products/100000422/wechsler-preschool-and-primary-scale-of-intelligence-third-edition-wpsi-iii.html
Wechsler Intelligence Scale for Children–IV (WISC-IV)	6–16 yrs	Gives an FSIQ score. 15 Subtests; 4 core areas: verbal comprehension, perceptual reasoning, working memory, and processing speed.	Available for purchase at: http://www.pearsonclinical.com/psychology/products/100000310/wechsler-intelligence-scale-for-children-fourth-edition-wisc-iv.html
Leiter International Performance Scale –3(Leiter –3)	3–75+ y	Give an NVIQ. Two groupings of subtests: cognitive and attention/memory, 10 subtests total. Measures fluid reasoning, visualization, memory, attention, cognitive interference	Available for purchase at: http://www.wpspublish.com/store/p/2840/leiter-international-performance-scale-third-edition-leiter-3
Mullen Scales of Early Learning	Birth to 68 mo	Measures cognitive ability and motor development; 5 scales: gross motor, visual reception, fine motor, expression language, receptive language	Available for purchase at: http://www.pearsonclinical.com/childhood/products/100000306/mullen-scales-of-early-learning.html

[https://www.jaacap.org/article/S0890-8567\(19\)32223-3/pdf](https://www.jaacap.org/article/S0890-8567(19)32223-3/pdf)

Common Measures of Adaptive Functioning in Children and Adolescents

Instrument and current version	Age Range	Description:	Availability
Vineland Adaptive Behavior Scales—II	Birth to adulthood	Survey Interview, Parent/Caregiver Rating Forms, Expanded Interview Form, Teacher Rating Form. Assesses communication, daily living skills, socialization, motor skills, and	Available for purchase at: http://www.pearsonclinical.com/psychology/products/100000668/vineland-adaptive-behavior-scales-second-edition-vineland-ii-
Scales of Independent Behavior—Revised	Infancy-80+ years	Interview or checklist. Assesses 14 areas of adaptive behavior and 8 areas of problem behavior.	Available for purchase at: http://www.hmhco.com/hmh-assessments/other-clinical-assessments/sib-r
AAIDD's Diagnostic Adaptive Behavior Scale (DABS)	4–21 y	Under development. Standardized assessment including three domains: conceptual skills, social skills, and practical skills.	Not yet available
Adaptive Behavior Assessment Scale (ABAS II)	Birth to 89 y	Parent/Primary Caregiver form or Teacher/daycare provider form. Assesses three domains of adaptive behavior: conceptual, social, and practical	Available for purchase at: http://www.pearsonclinical.com/psychology/products/100000449/adaptive-behavior-assessment-system-second-edition-abas-second-edition.html

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Inappropriate, life-threatening or challenging behaviors may be inadvertently maintained by reinforcement from others in the environment. Which of the following is a process that can be carried out in order to help identify the factors maintaining the behavior?

- a) Functional analysis
- b) Statistical analysis
- c) Behavioral analysis
- d) Procedural analysis

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Functional Analysis: The use of operant conditioning principles used to try and understand what rewarding or reinforcing factors might be maintaining behavior



Which of the following is a form of child abuse that is known to cause intellectual disability?

- a) Shaken baby syndrome
- b) Abused child syndrome
- c) Battered baby syndrome
- d) Damaged infant syndrome

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- a) 45-50
- b) 55-60
- c) 35-55.
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Disorders which are characterized by serious abnormalities in the developmental process, fall under the heading of pervasive developmental disorders (PDD) and are usually associated with impairment in several areas of development. From early infancy, some children will exhibit a spectrum of developmental impairments and delays that include which of the following?

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- b) Intellectual disabilities
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Pervasive Developmental Disorders (PDDs): A group of disorders which are characterised by serious abnormalities in the developmental process, usually associated with impairment in several areas of development. The most commonly diagnosed PDDs are Autistic Disorder (Autism), Rett's Disorder, Childhood Disintegrative Disorder, and Asperger's Syndrome



Stacy has an intellectual disability (ID). She had delays sitting up and walking as an infant and has been diagnosed with a biological anomaly, though now, at age 6, she can use the toilet on her own. Which degree of intellectual disability does Stacy most likely have?

- A. mild
- B. moderate
- C. severe
- D. profound

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Which of the following are domains of adaptive functioning?

- A. conceptual, social, practical
- B. social, emotional, behavioral
- C. academic and cognitive
- D. eating, sleeping, playing

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Which of the following is true of conditions that cause intellectual disability (ID)?

- A. Over 500 unique causes of ID have been identified.
- B. Genetic mutation is a category of causes that result in ID.
- C. Known conditions explain nearly 100% of cases of ID.
- D. Down syndrome is a chromosomal abnormality that can cause ID.

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How is adaptive functioning typically assessed?

- A. via the child's self-report
- B. via an intelligence test
- C. via a checklist that parents fill out
- D. via a semi structured interview given to caregivers

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Which of the following is true of age and intellectual disability (ID)?

- A. Only children can be diagnosed with ID.
- B. In order to be diagnosed with ID, onset of the intellectual and adaptive functioning problem must be in early childhood.
- C. There is no age criterion for ID; anyone could develop difficulties at any age and be diagnosed with the condition.
- D. ID is continuous with Alzheimer's dementia; when diagnosed in those above age 18, the diagnosis would be Alzheimer's dementia.



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Which of the following is a similarity between Down syndrome and Prader-Willi syndrome?

- A. Both are genetic disorders.
- B. Both relate to missing paternal genetic information.
- C. Both result in relative strengths in communication skills.
- D. Neither are associated with reduced life expectancy, compared to typically developing individuals.



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Hyperphagia is associated with which condition?

- A. Down syndrome
- B. Angelman syndrome
- C. Williams syndrome
- D. Prader-Willi syndrome

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Which of the following is associated with Prader-Willi syndrome?

- A. hoarding of food
- B. anxiety
- C. high IQ
- D. depression

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Which of the following conditions are associated with a happy demeanor?

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- C. Angelman syndrome
- D. Williams syndrome

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B. Prader-Willi syndrome

C. Angelman syndrome

D. Williams syndrome

The End

